

Validation and Optimization

Series: M2S | Notebook: 8 of 8 | Created: January 2026

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Prerequisites

Before starting this notebook, you should have:

Requirement	Description
Completed M2S-01 to M2S-07	All migration steps executed
Agents migrated	OneAgents reporting to SaaS
Configurations applied	Settings imported to SaaS
Integrations configured	Notifications working

Learning Objectives

By the end of this notebook, you will:

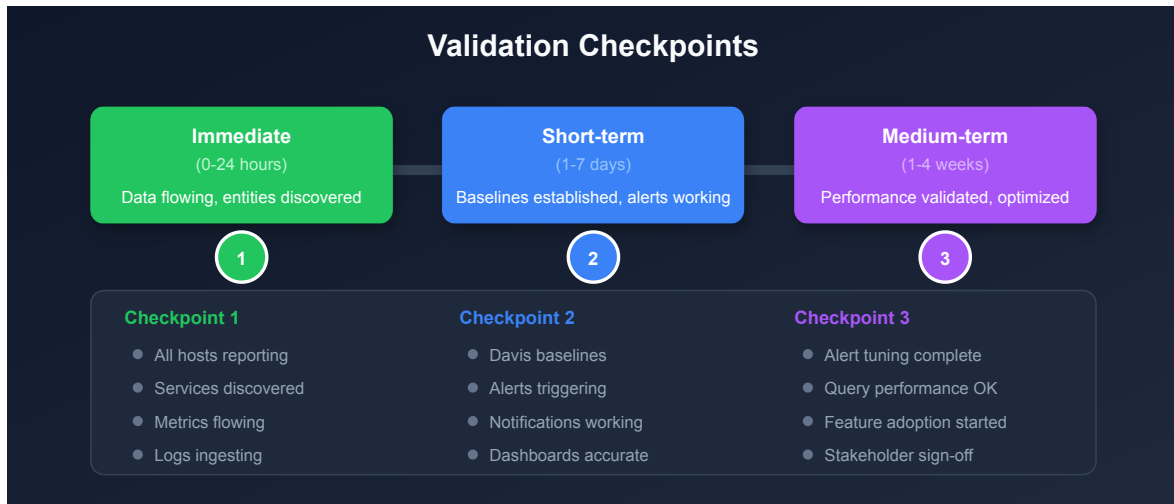
- Validate complete migration success
 - Verify data quality and completeness
 - Optimize performance and configuration
 - Plan adoption of new SaaS features
 - Complete the migration formally
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1. Introduction

Validation is the final and most critical phase. A migration isn't complete until all success criteria are verified and stakeholders sign off.

Validation Approach

Phase	Focus
Immediate (0-24h)	Data flowing, entities discovered
Short-term (1-7d)	Baselines established, alerts working
Medium-term (1-4w)	Performance validated, optimization complete



2. Migration Validation

2.1 Entity Count Validation

Compare entity counts between Managed (documented during planning) and SaaS:

```
// Complete entity summary
fetch dt.entity.host | summarize hosts = count()
| append [fetch dt.entity.service | summarize services = count()]
| append [fetch dt.entity.application | summarize applications = count()]
| append [fetch dt.entity.process_group | summarize processGroups = count()]
| append [fetch dt.entity.synthetic_test | summarize syntheticMonitors = count()]
```

2.2 Host Validation

Verify all expected hosts are reporting:

```
// List all hosts (to verify reporting)
fetch dt.entity.host
| fieldsAdd hostName = entity.name
| fields hostName, id
| sort hostName desc
| limit 50
```

```
// Check host activity by looking at recent metrics
// Hosts not sending metrics may have issues
timeseries avgCpu = avg(dt.host.cpu.usage), by:{dt.entity.host}
| filter isNull(avgCpu)
| fields dt.entity.host
```

2.3 Service Discovery Validation

```
// Count services by technology
fetch dt.entity.service
| summarize count = count(), by:{serviceType}
| sort count desc
```

```
// Count active services
fetch dt.entity.service
| summarize activeServices = count()
```

2.4 ActiveGate Validation

```
// ActiveGate validation – use the Entities API v2 or Settings API
// Note: ActiveGates are not directly queryable via DQL fetch
// Use: GET /api/v2/entities?entitySelector=type("ENVIRONMENT_ACTIVE_GATE")
// Or check the ActiveGates page in the Dynatrace UI
```

2.5 Validation Comparison Table

Entity Type	Expected (from Planning)	Actual (SaaS)	Status
Hosts	—	—	[] Match
Services	—	—	[] Match
Applications	—	—	[] Match
Process Groups	—	—	[] Match
Synthetic Monitors	—	—	[] Match
ActiveGates	—	—	[] Match

3. Data Quality Checks

3.1 Metrics Flow Validation

```
// Check CPU metrics are flowing
timeseries avg(dt.host.cpu.usage), by:{dt.entity.host}
| limit 10
```

```
// Check memory metrics
timeseries avg(dt.host.memory.usage), by:{dt.entity.host}
| limit 10
```

3.2 Log Ingestion Validation

```
// Check log volume over time
fetch logs
| summarize logCount = count(), by:{time_bucket = bin(timestamp, 5m)}
| sort time_bucket desc
| limit 24
```

```
// Check logs by source
fetch logs
| summarize count = count(), by:{log.source}
| sort count desc
| limit 10
```

3.3 Trace/Span Validation

```
// Check distributed traces are flowing
fetch spans
| summarize spanCount = count(), by:{time_bucket = bin(timestamp, 5m)}
| sort time_bucket desc
| limit 12
```

```
// Check service request volume
fetch spans
| filter span.kind == "server"
| summarize requestCount = count(), by:{dt.entity.service}
| sort requestCount desc
| limit 10
```

3.4 Data Gaps Detection

```
// Check for metric data continuity over time
// Use timeseries to query metrics – count hosts with data per time bucket
timeseries avgCpu = avg(dt.host.cpu.usage), by:{dt.entity.host}
| summarize hostsWithData = count()
| limit 1
```

4. Performance Optimization

4.1 Query Performance

Review and optimize frequently-used queries:

Optimization	Technique
Add time filters	Always filter by timestamp
Limit results	Use <code>limit</code> to reduce data
Filter early	Apply filters before aggregations
Use summarize	Aggregate instead of fetching all rows

4.2 Dashboard Optimization

Issue	Solution
Slow loading	Reduce tile count, optimize queries
Timeouts	Add time filters, reduce scope
Too much data	Use summarize/aggregations

4.3 Alert Tuning

Symptom	Action
Too many alerts	Raise thresholds, add filters
Missing alerts	Lower thresholds, verify coverage
Noisy alerts	Add time-based conditions

4.4 Baseline Establishment

Davis AI needs time to establish baselines:

Baseline Type	Time Required
Response time	2-7 days
Error rates	2-7 days
Resource usage	7-14 days
Traffic patterns	7-14 days

Note: Expect more alerts initially as baselines establish. This is normal.

5. Feature Adoption

5.1 SaaS-Only Features to Explore

Feature	Benefit	Priority
Grail	Unified data lakehouse	High
Notebooks	Interactive analysis	High
Automations	Advanced workflows	Medium
Platform Apps	Extended functionality	Medium
Davis Copilot	AI-assisted analysis	High

5.2 Grail Adoption

Grail provides unified querying across all data:

Data Type	Grail Table
Logs	<code>fetch logs</code>
Spans	<code>fetch spans</code>
Events	<code>fetch events</code>
Business Events	<code>fetch bizevents</code>
Entities	<code>fetch dt.entity.*</code>

5.3 Automations Adoption

Replace manual processes with workflows:

Use Case	Workflow Type
Problem remediation	Problem trigger
Report generation	Scheduled trigger
Event response	Event trigger
Integration updates	HTTP request action

5.4 Feature Adoption Plan

Week	Focus
1-2	Stabilize migration, tune alerts
3-4	Explore Notebooks, create team notebooks
5-6	Implement first automation workflows
7-8	Adopt platform apps, explore Copilot

6. Migration Completion

6.1 Final Validation Checklist

Category	Checkpoint	Status
Entities	All hosts reporting	☐
Entities	All services discovered	☐
Entities	All applications configured	☐
Data	Metrics flowing	☐
Data	Logs ingesting	☐
Data	Traces capturing	☐
Config	Dashboards working	☐
Config	Alerting functional	☐
Config	Notifications delivering	☐
Security	SSO configured	☐
Security	Access controls applied	☐
Integration	All webhooks working	☐

6.2 Stakeholder Sign-Off

Stakeholder	Approval	Date
Platform Team Lead	☐	
Security Team	☐	
Application Teams	☐	
Executive Sponsor	☐	

6.3 Managed Decommissioning

After successful validation:

Step	Action	Timing
1	Confirm no agents pointing to Managed	Immediate
2	Keep Managed for comparison	2-4 weeks
3	Export final Managed data	Before shutdown
4	Decommission Managed cluster	After validation period

6.4 Documentation Updates

Update your documentation:

Document	Update Required
Runbooks	New SaaS URLs and procedures
Architecture diagrams	New network topology
Onboarding guides	SaaS-specific instructions
Disaster recovery	Updated recovery procedures

Migration Success Metrics

Metric	Target	Actual
Host coverage	100%	
Service coverage	100%	
Data gaps	< 15 minutes	
Alert delivery	100%	
Dashboard availability	100%	
Integration success	100%	

Series Conclusion

Congratulations on completing the M2S migration series!

What You've Accomplished

Notebook	Achievement
M2S-01	Understood benefits and motivation
M2S-02	Learned the migration framework
M2S-03	Completed planning and assessment
M2S-04	Designed architecture and network
M2S-05	Migrated configurations
M2S-06	Migrated agents
M2S-07	Addressed security and privacy
M2S-08	Validated and optimized

Ongoing Success

Your migration is complete, but the journey continues:

1. **Monitor baselines** - Watch Davis AI establish patterns
2. **Tune alerting** - Reduce noise over time
3. **Adopt features** - Leverage SaaS capabilities
4. **Share knowledge** - Train your teams on SaaS
5. **Provide feedback** - Help improve the platform

Additional Resources

- [Dynatrace Documentation](#)
 - [Dynatrace Community](#)
 - [Dynatrace University](#)
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Summary

In this final notebook, you learned:

- How to validate migration completeness
- DQL queries for data quality verification
- Performance optimization techniques
- SaaS feature adoption planning
- Migration completion procedures

Key Takeaway: A migration is only complete when all success criteria are validated and stakeholders sign off. Take time to verify everything before decommissioning Managed.

Thank you for completing the M2S: Managed-to-SaaS Migration Best Practices series!

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